

LEARNING APP: MULTI-PLATFORM EDUCATIONAL INFRASTRUCTURE – TEAM 3

This project presents a scalable educational platform ecosystem composed of a centralized backend API, a Flutter-based mobile application, and a React-based administration dashboard. The system is designed using a modular architecture that separates business logic, client applications, and administrative operations into independent but interconnected repositories.

The primary objective of the project is to build a production-oriented learning infrastructure that supports secure authentication, progress tracking, modular content delivery, quiz management, and scalable system design.

Group Members:

- **Ecem Nur Özen:** Responsible for the complete technical development of the project ecosystem, including backend API architecture, database design, authentication system, mobile application development, admin dashboard development, system integration, software engineering processes, and overall infrastructure implementation.
- **Karya Kinyas:** Responsible for creating and organizing educational content within the platform, including lesson structures, topic flows, and quiz materials through the admin dashboard.
- **Betül Doğan:** Responsible for preparing and managing learning materials, educational resources, and platform content organization using the admin management system.
- **Gizem Gürtürk:** Responsible for educational content contribution, lesson planning, quiz preparation, and maintaining instructional materials through the admin panel.

1. INTRODUCTION

Modern educational platforms require scalable infrastructures capable of serving multiple client applications while maintaining centralized business logic and secure data management. Traditional monolithic systems often struggle with scalability, maintainability, and deployment flexibility.

This project addresses these challenges by implementing a modular fullstack ecosystem where the backend API, mobile client, and admin dashboard operate as separate layers connected through RESTful communication.

2. PROBLEM DEFINITION

Many educational applications are developed without clear architectural separation, leading to long-term maintainability and scalability problems.

Common challenges include:

- Monolithic application structures
- Tight frontend-backend coupling
- Difficult deployment workflows
- Limited scalability
- Lack of centralized authentication
- Weak administrative tooling

3. PROPOSED SOLUTION

The proposed solution is a multi-platform ecosystem built around a centralized ASP.NET Core backend API.

The ecosystem consists of:

- Backend API for business logic and data services
- Flutter mobile application for end users
- React admin dashboard for management operations

Each component is independently scalable while communicating through standardized REST APIs.

4. SYSTEM ARCHITECTURE

The backend is implemented using ASP.NET Core Web API with a layered architecture approach.

PostgreSQL is used as the relational database system for persistent storage.

Flutter enables cross-platform mobile deployment while React powers the administrative dashboard.

JWT authentication is used across the ecosystem for secure communication.

5. TECHNICAL MILESTONES

Backend Milestones:

- ASP.NET Core API initialization
- PostgreSQL integration
- JWT authentication infrastructure
- Entity Framework migrations
- REST API implementation

Mobile Milestones:

- Flutter project setup
- Authentication screens
- Navigation architecture
- Quiz and progress interfaces

Dashboard Milestones:

- React setup
- Protected route system
- CRUD management pages
- API integration layer

6. DATABASE DESIGN APPROACH

The database layer follows relational modeling principles.

Core entities include Users, Topics, Lessons, Quizzes, Quiz Questions, User Progress, and Achievements.

The schema is designed to support scalability, analytics, and future feature expansion.

7. FUTURE DEVELOPMENT TARGETS

Planned improvements include:

- Docker containerization
- CI/CD automation
- AI-assisted recommendation systems
- Push notifications
- Analytics dashboards
- Real-time monitoring systems

8. WHY THIS PROJECT MATTERS

This project demonstrates how modern software engineering principles can be applied to educational technology systems.

The ecosystem emphasizes scalable architecture, separation of concerns, secure authentication, maintainable code organization, and production-oriented infrastructure planning.

9. CONCLUSION

This project presents a modular learning platform ecosystem integrating a centralized backend API, Flutter mobile application, and React administration dashboard.

The ecosystem demonstrates practical implementation of scalable fullstack software engineering principles.

Github Link: [LearningApp_ComNet](#) (This repo contains 3 different repositories. Links to other repositories are given in this main repository's Readme file.)